

From Points to Prints

Generating Building Roofprints and Footprints from Airborne Lidar Data and Inaccurate Outlines

A4 Presentation

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Context

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Polygon deformation	8
Roofprints	9
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Roofprint vs. footprint

	Roofprint	Footprint
Defined by	Roof	Façades
Comes from	Aerial data (images, ALS)	Terrain measurements (TLS, MLS)
Used for	3D modelling, solar potential	Tax estimation, energy consumption

Table 1: Comparison between roofprints and footprints

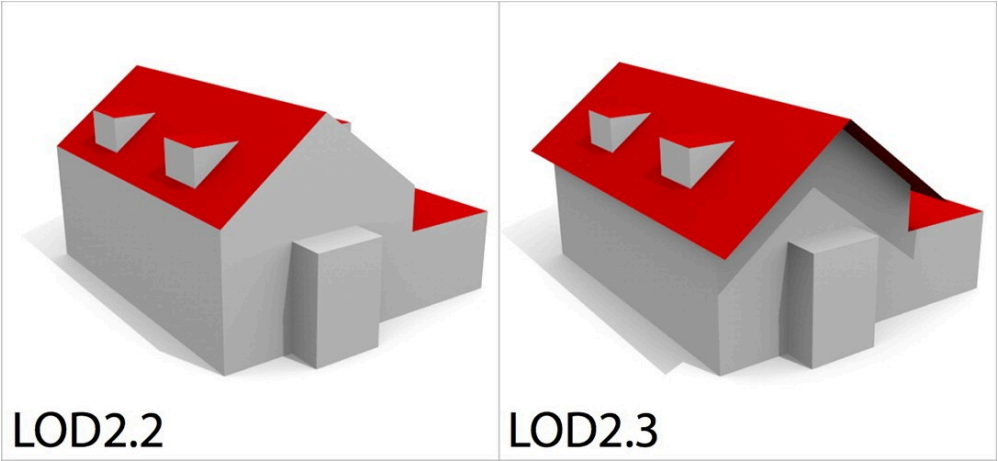


Figure 1: Part of the visual definition of buildings Level of Detail (LoD) containing LoD 2.2 and 2.3 (Biljecki et al., 2016) .

Strengths and weaknesses of BD TOPO

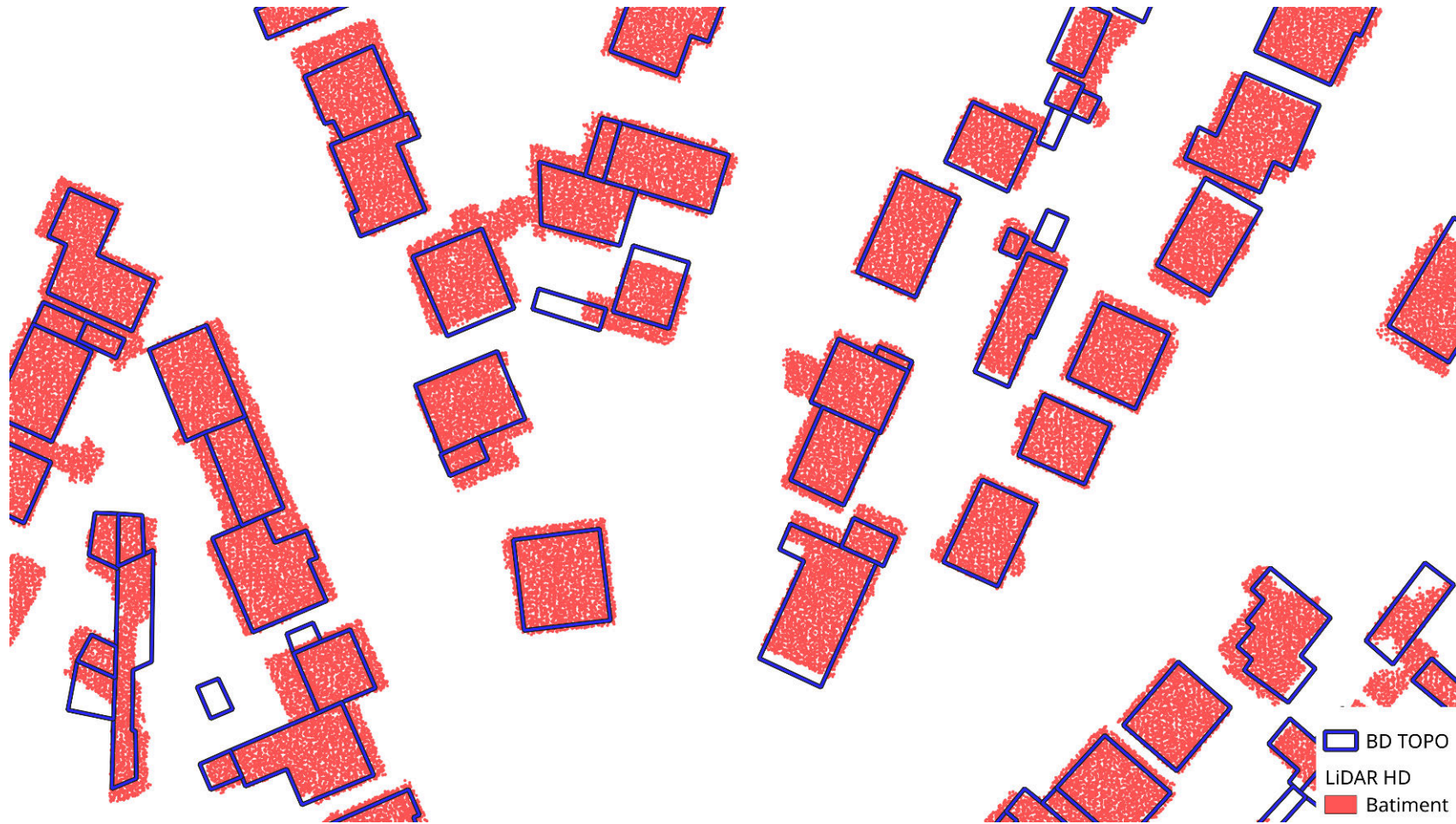
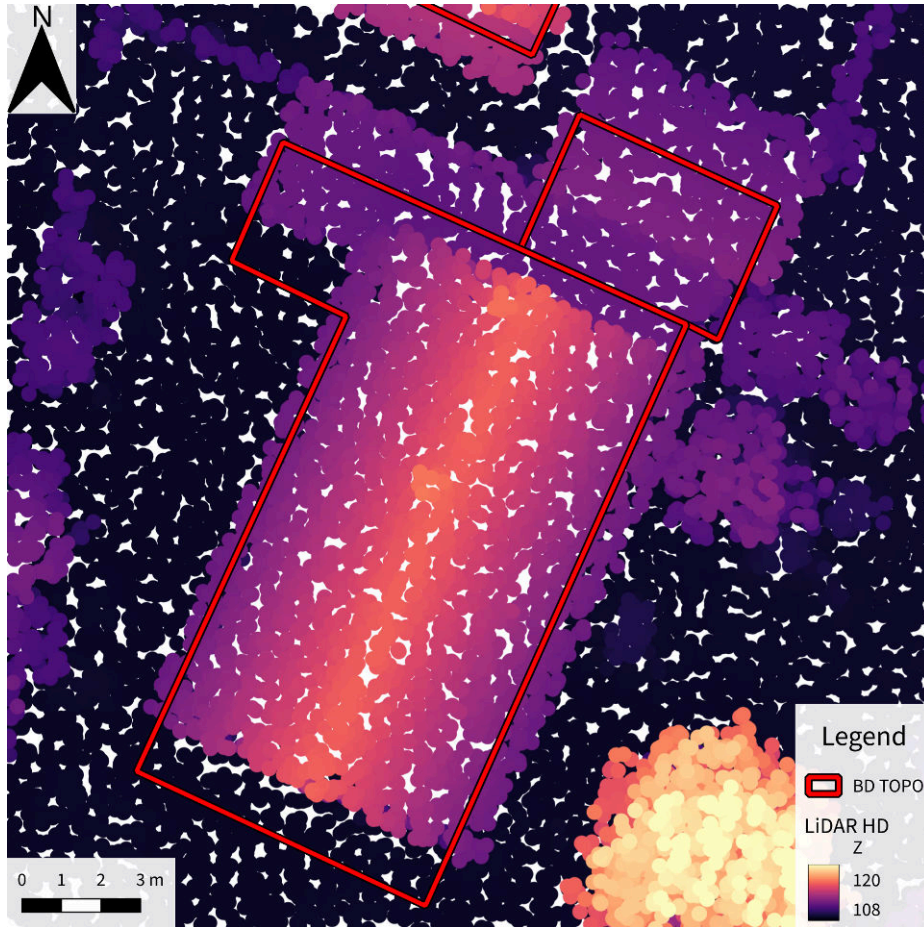


Figure 2: BD TOPO buildings are most of the time footprints, and often shifted.

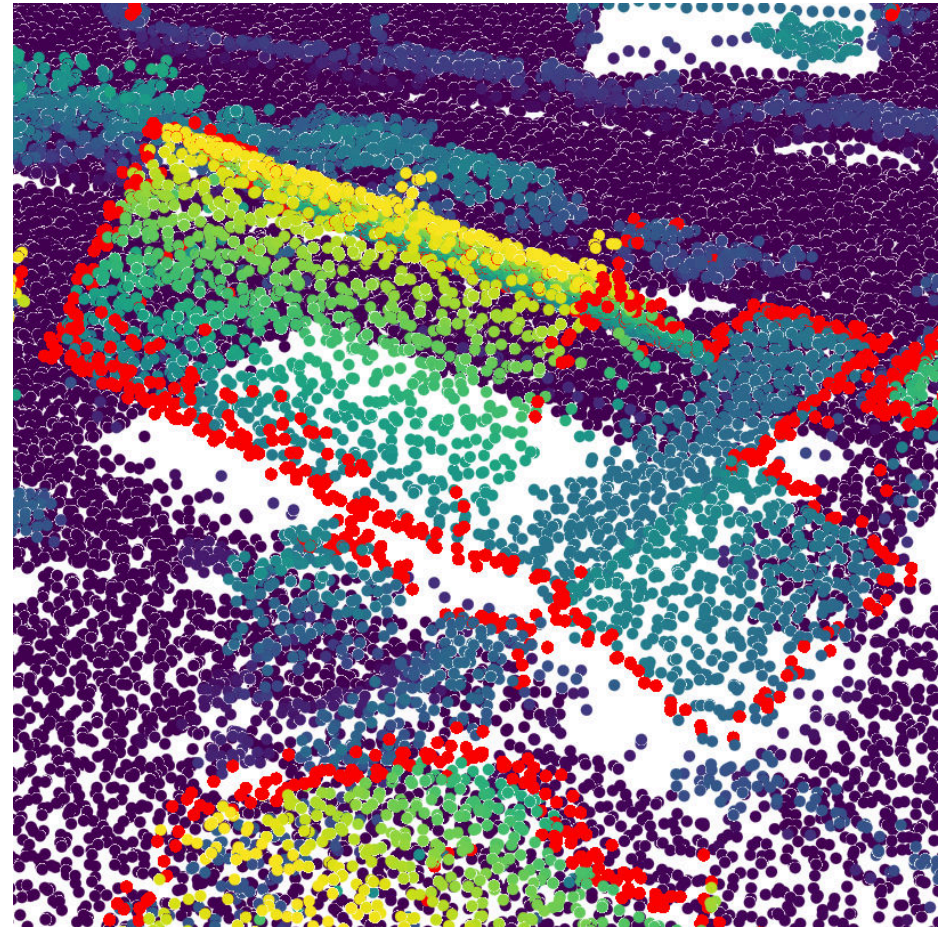
*How to generate coherent building **roofprints** and **footprints** from high-density **ALS** point clouds and existing **imprecise outlines**?*

Overview of the methodology

1. BD TOPO and LiDAR HD

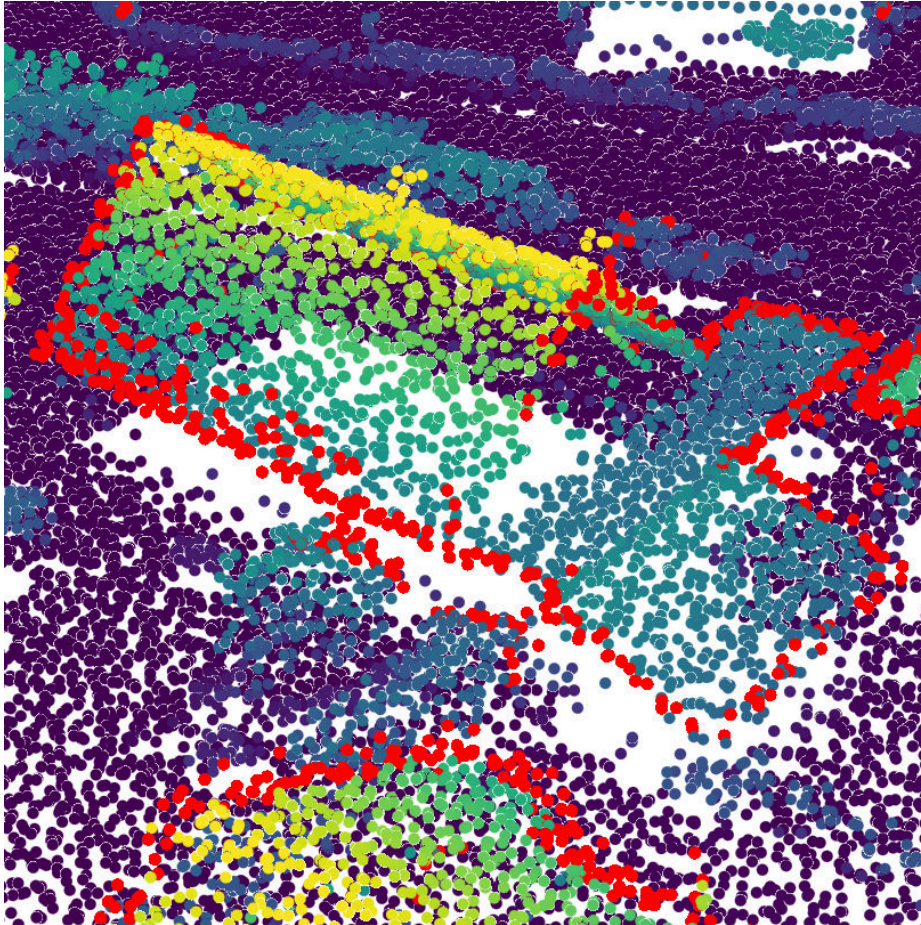


2. Find roof edge points

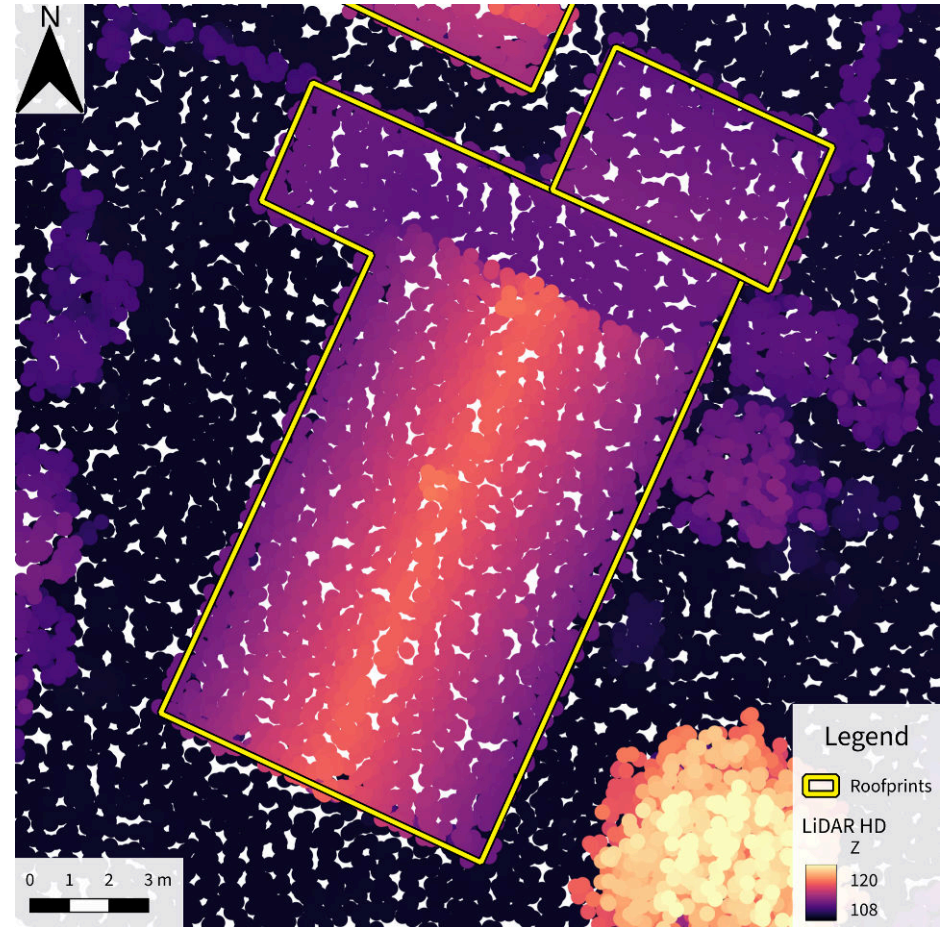


Overview of the methodology

2. Find roof edge points

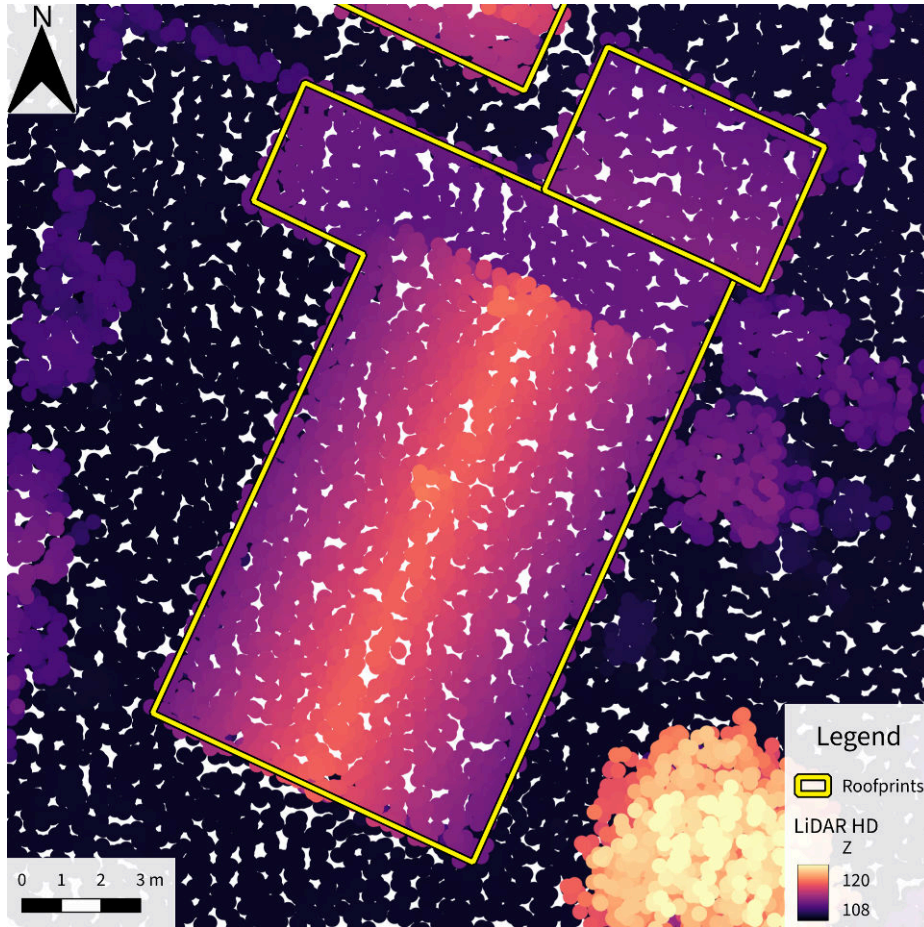


3. Align into a roofprint

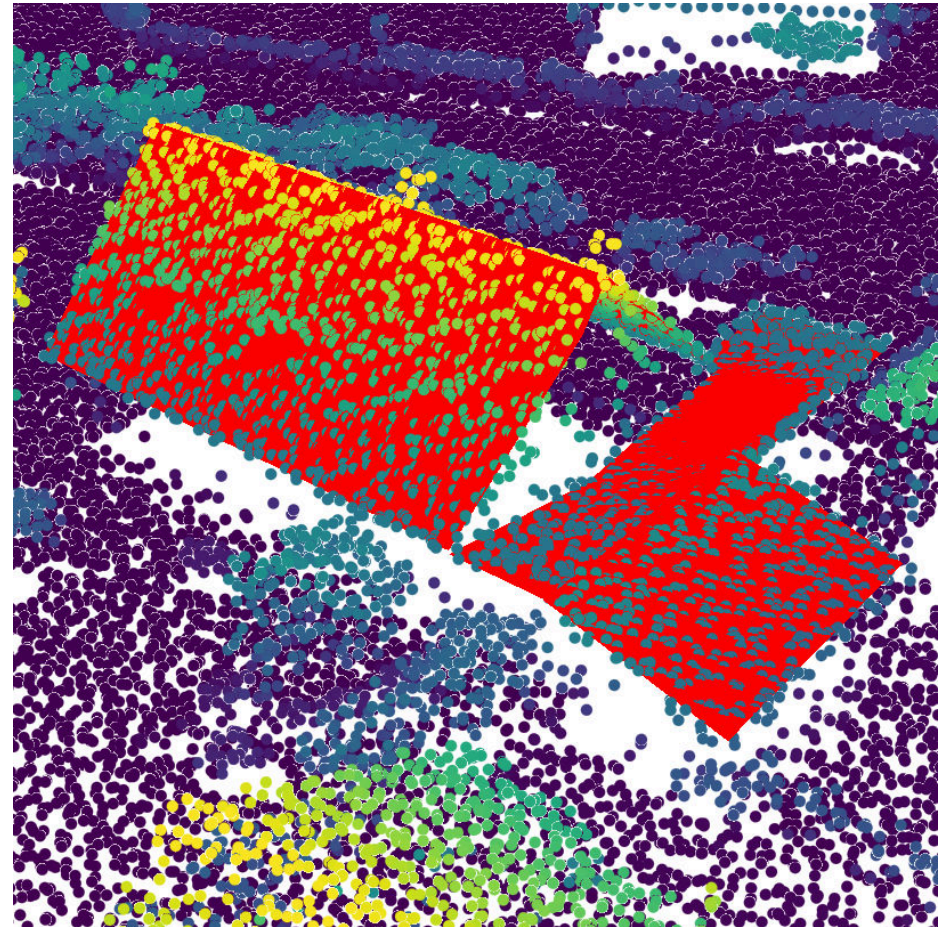


Overview of the methodology

3. Align into a roofprint

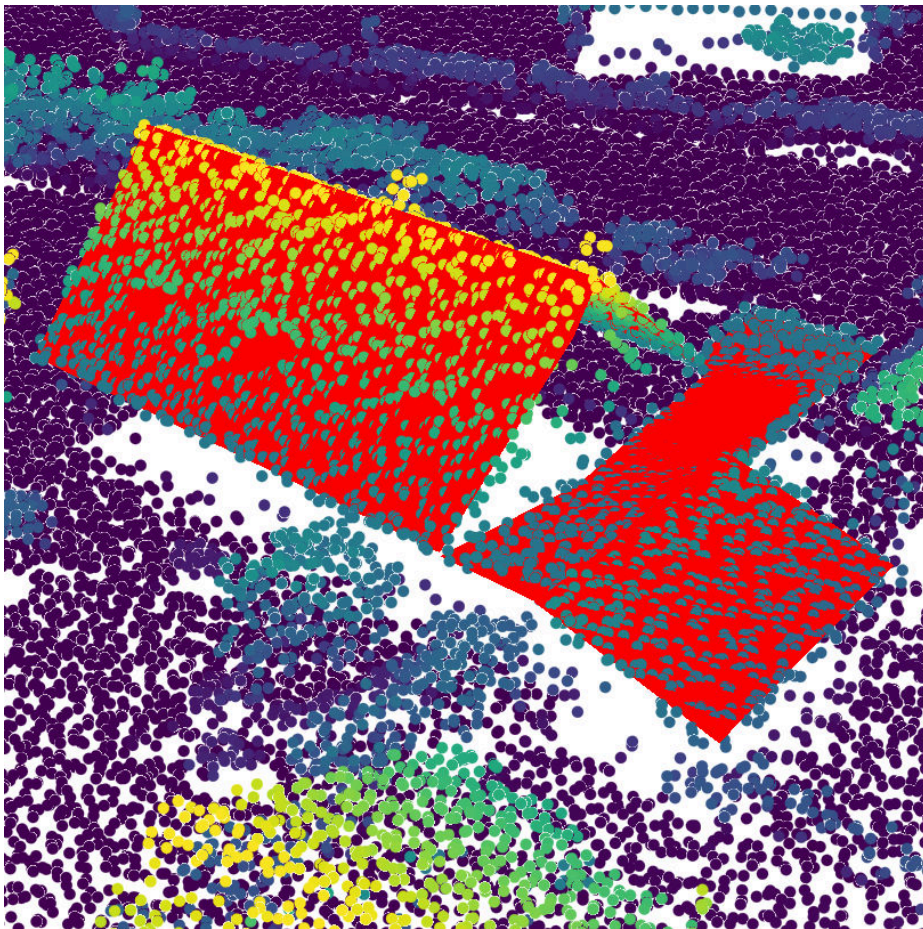


4. Generate 3D roof model

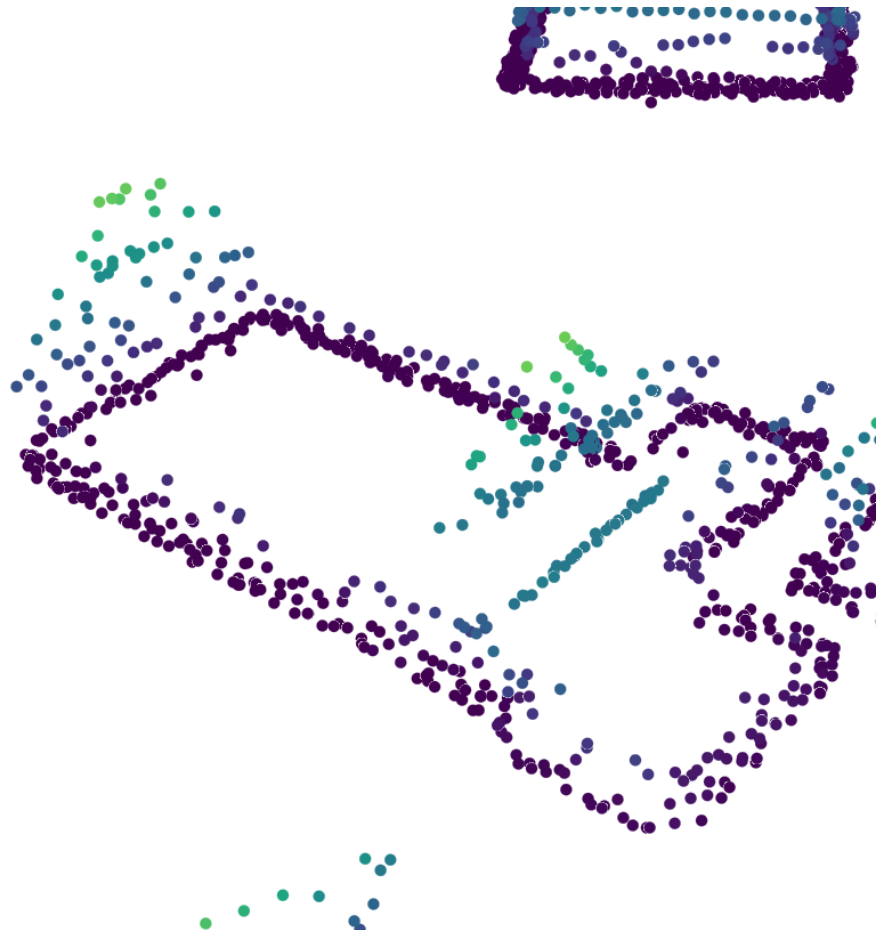


Overview of the methodology

4. Generate 3D roof model

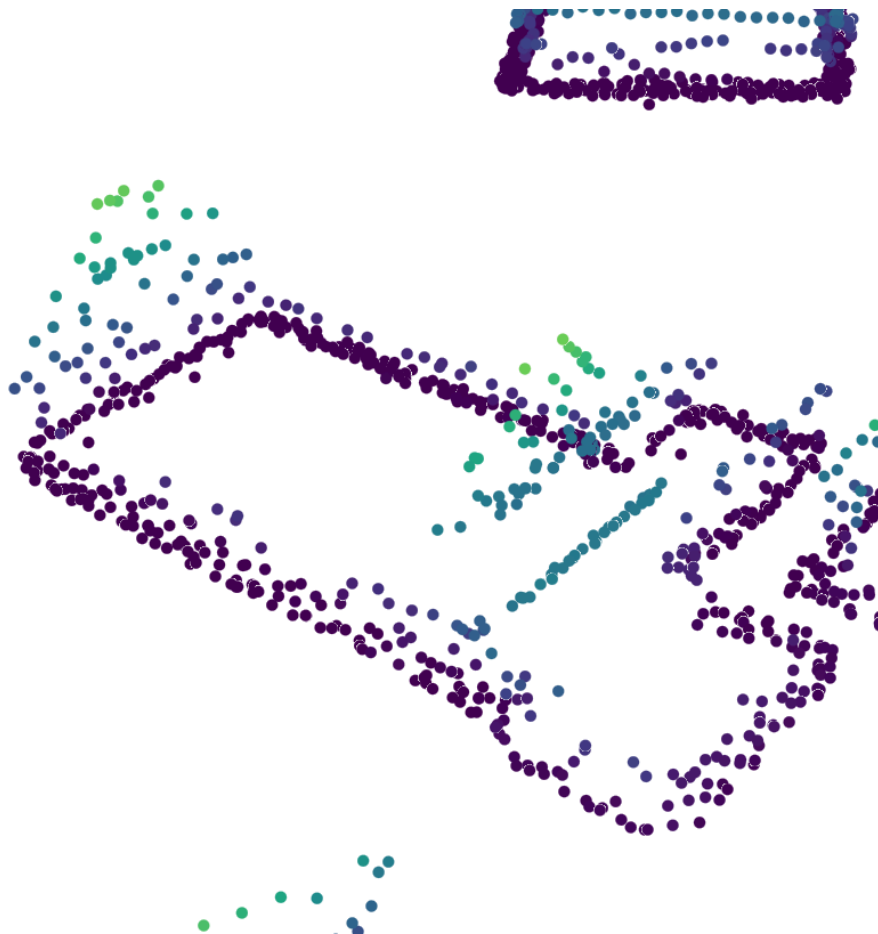


5. Find façade and ground points

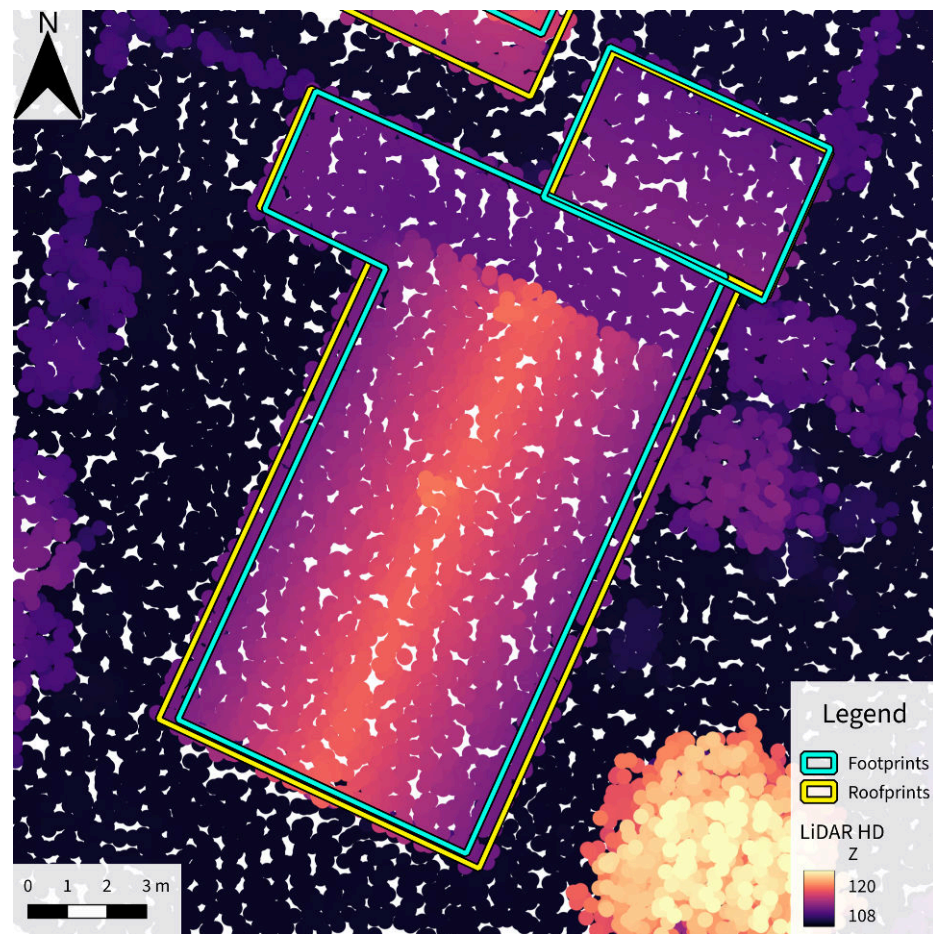


Overview of the methodology

5. Find façade and ground points



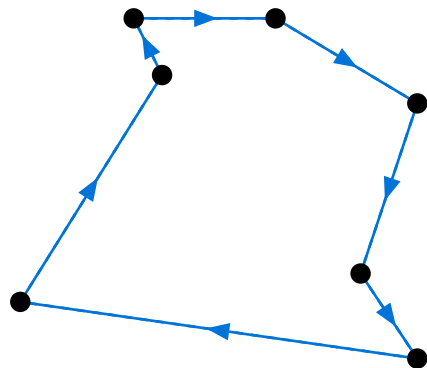
6. Align into a footprint



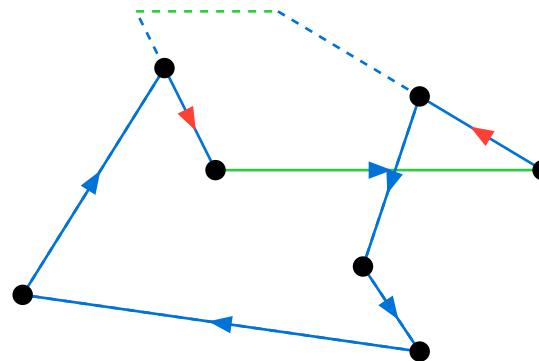
Polygon deformation

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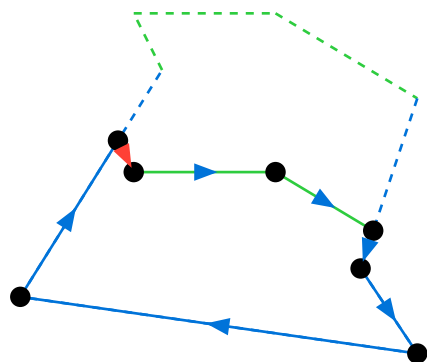
Our simple approach



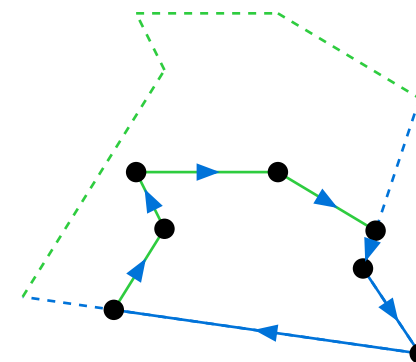
(a) Initial state.



(b) After shifting the focus edge.



(c) First step of the resolution.



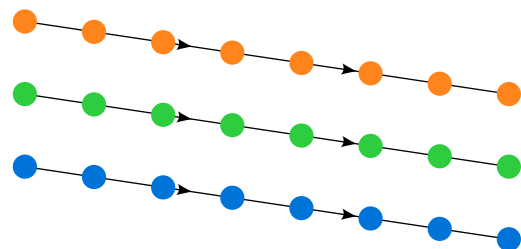
(d) Second and final step of the resolution.

Figure 3: Illustration of the polygon deformation algorithm on a single polygon.

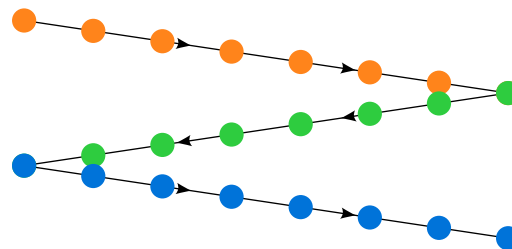
Roofprints

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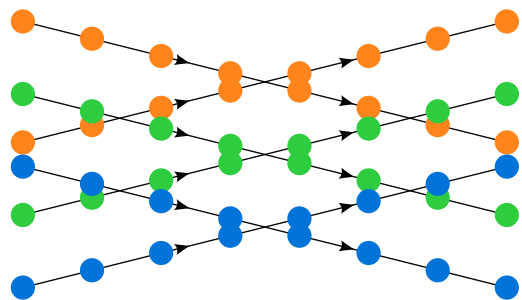
Point cloud topology



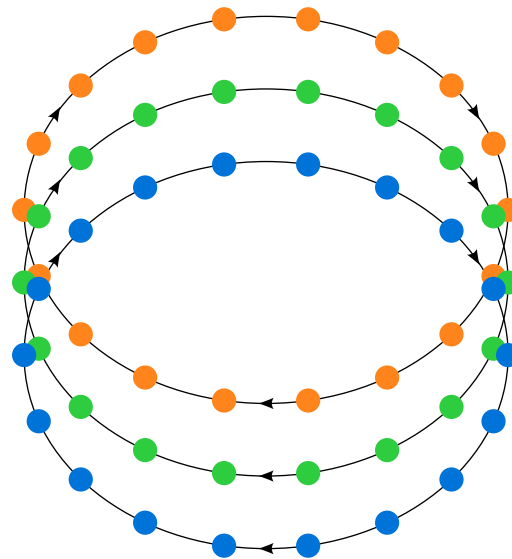
(a) Planar scan.



(b) Sine-wave scan.



(c) Cross scan.



(d) Circular scan.

Figure 4: Illustration in 2D (view from the top) of different types of ALS sensors. inspired from (Wu et al., 2026) .

Roofprints evidence

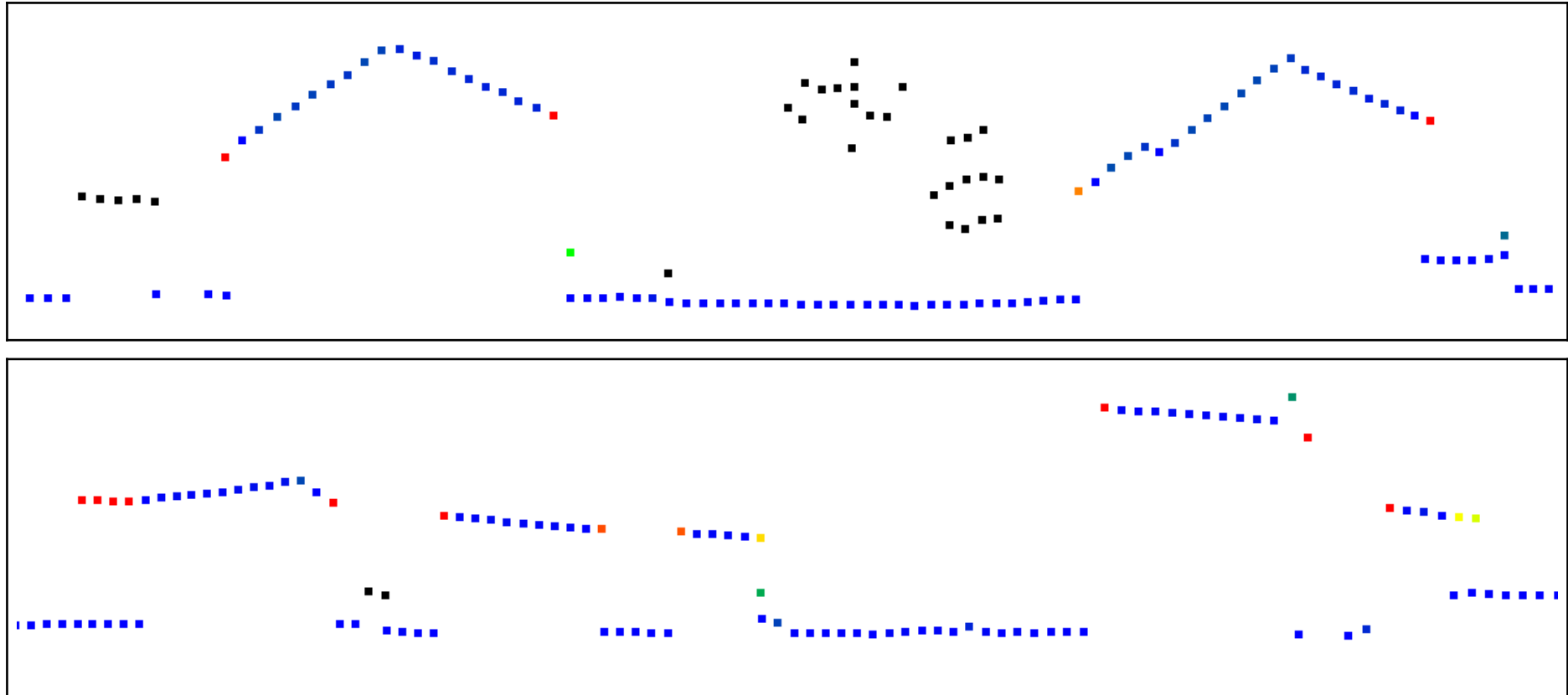


Figure 5: Two examples of the height differences obtained on a scan line. Black points are vegetation points, and for the other points the colour represents the value of Δh , with blue-green-yellow-red from low to high values.

Assign a score to roofprints

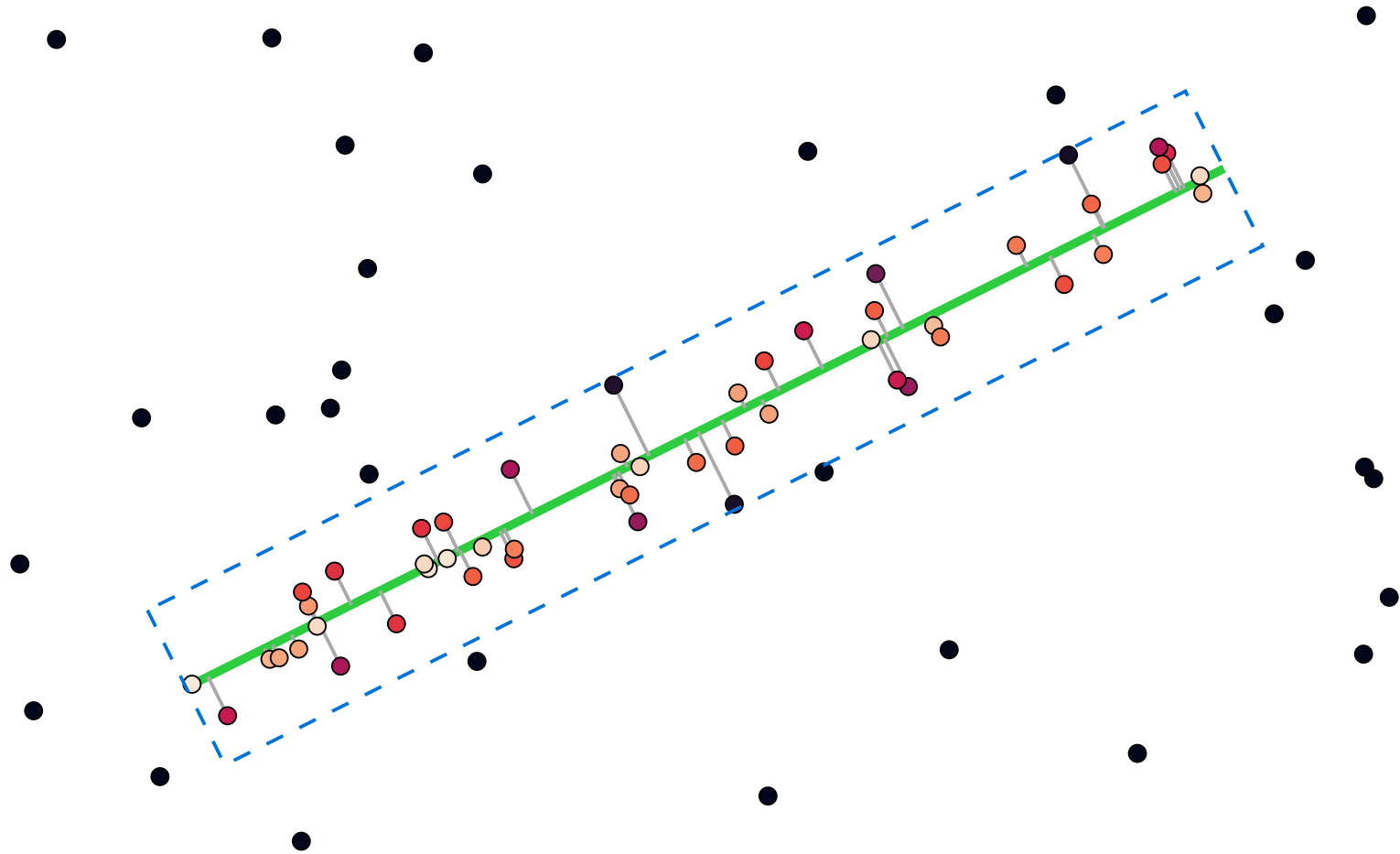
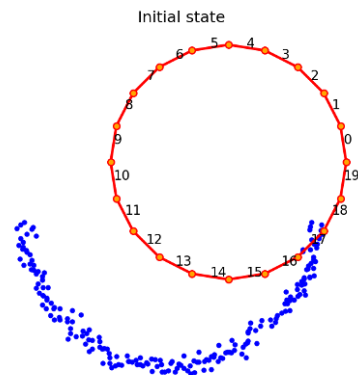


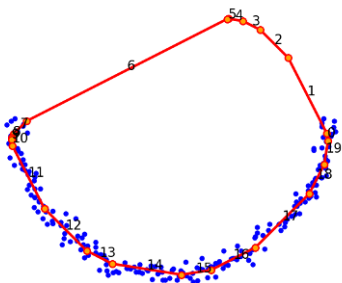
Figure 6: Illustration in 2D (view from the top) of the proximity score for an edge.

Need for regularization



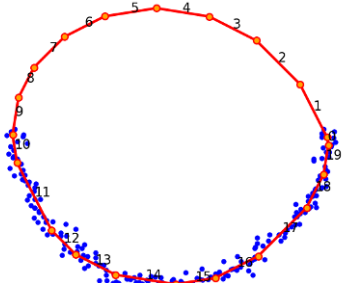
(a) Initial state

Step 5, shifted by 0.00000



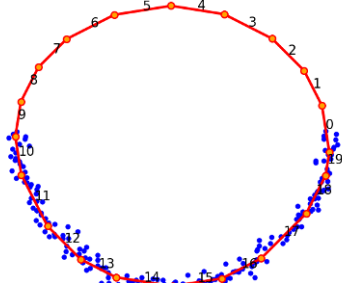
(b) $\alpha = 0.00$ (no regularisation)

Step 6, shifted by 0.00000



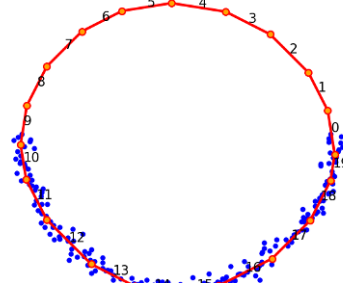
(c) $\alpha = 0.05$

Step 8, shifted by 0.00000



(d) $\alpha = 0.20$

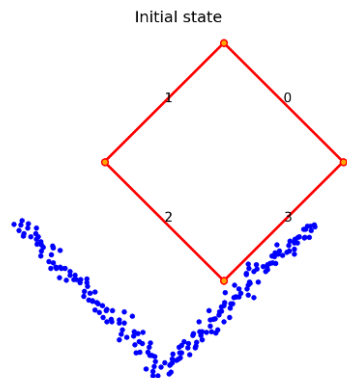
Step 7, shifted by 0.00000



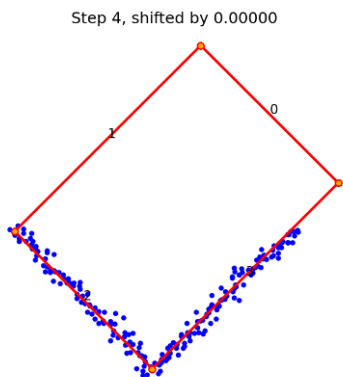
(e) $\alpha = 1.00$

Figure 7: Matching a circle to a half-circle of points with different values of the regularization parameter α .

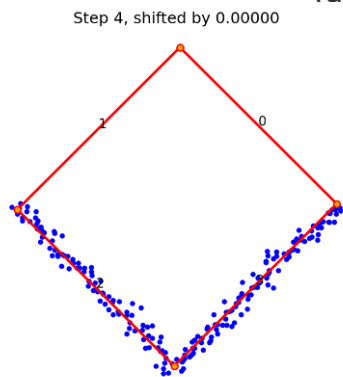
Need for regularization



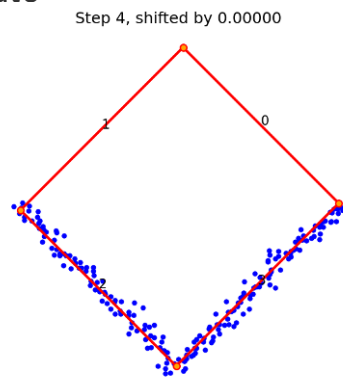
(a) Initial state



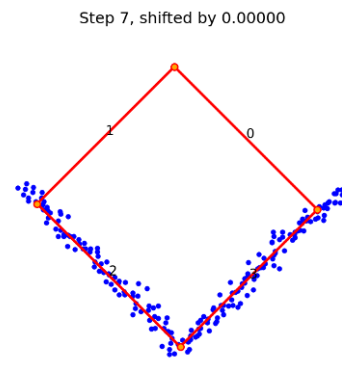
(b) $\alpha = 0.00$ (no regularization)



(c) $\alpha = 0.05$



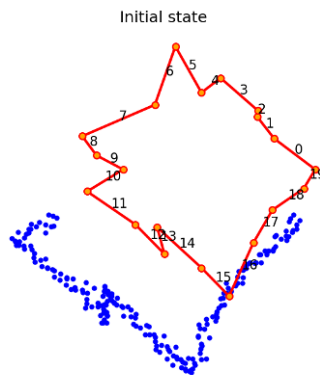
(d) $\alpha = 0.20$



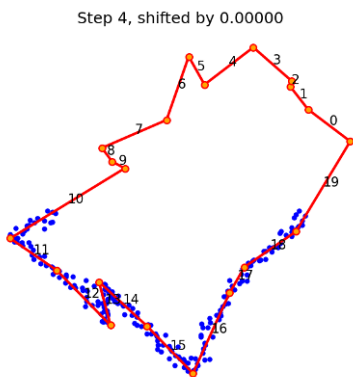
(e) $\alpha = 1.00$

Figure 8: Matching a square to points along two sides of a larger square with different values of the regularization parameter α .

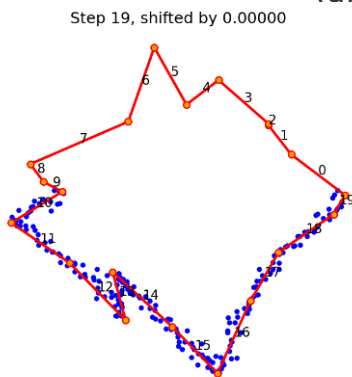
Need for regularization



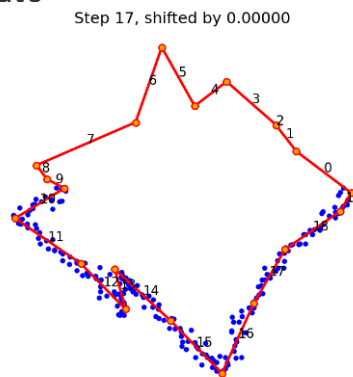
(a) Initial state



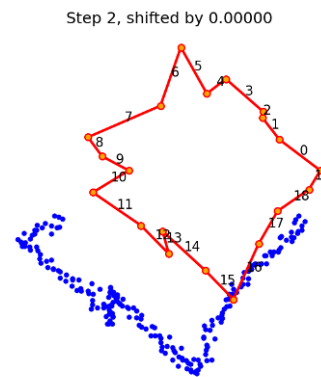
(b) $\alpha = 0.00$ (no regularization)



(c) $\alpha = 0.05$



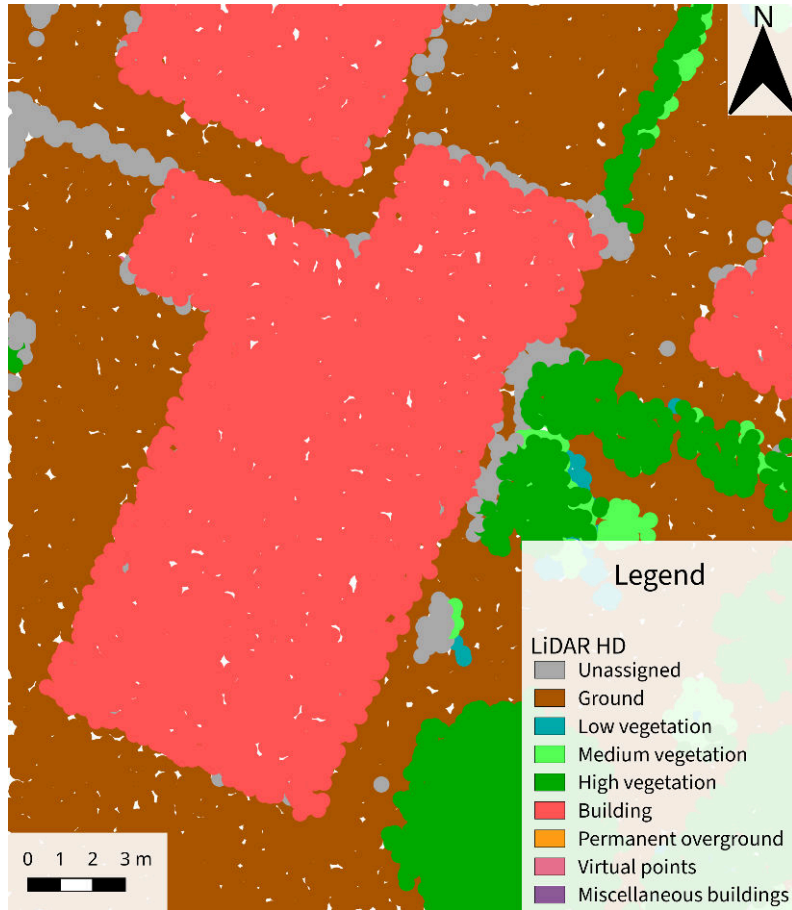
(d) $\alpha = 0.20$



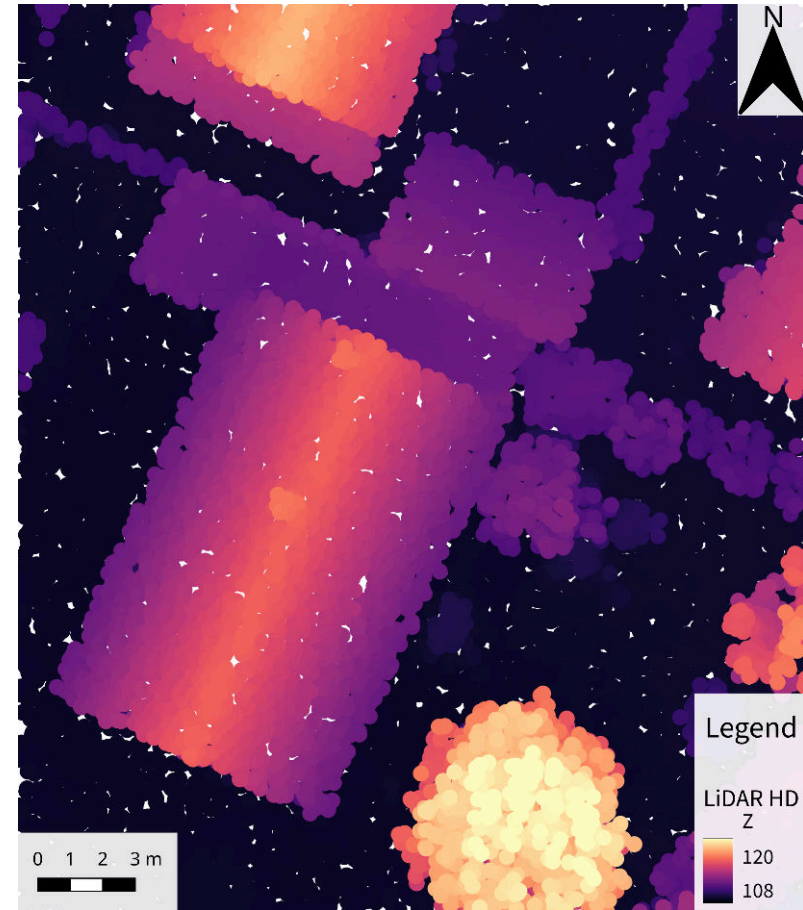
(e) $\alpha = 1.00$

Figure 9: Matching a polygon to half of its shape with points with different values of the regularization parameter α .

Illustration with a real building



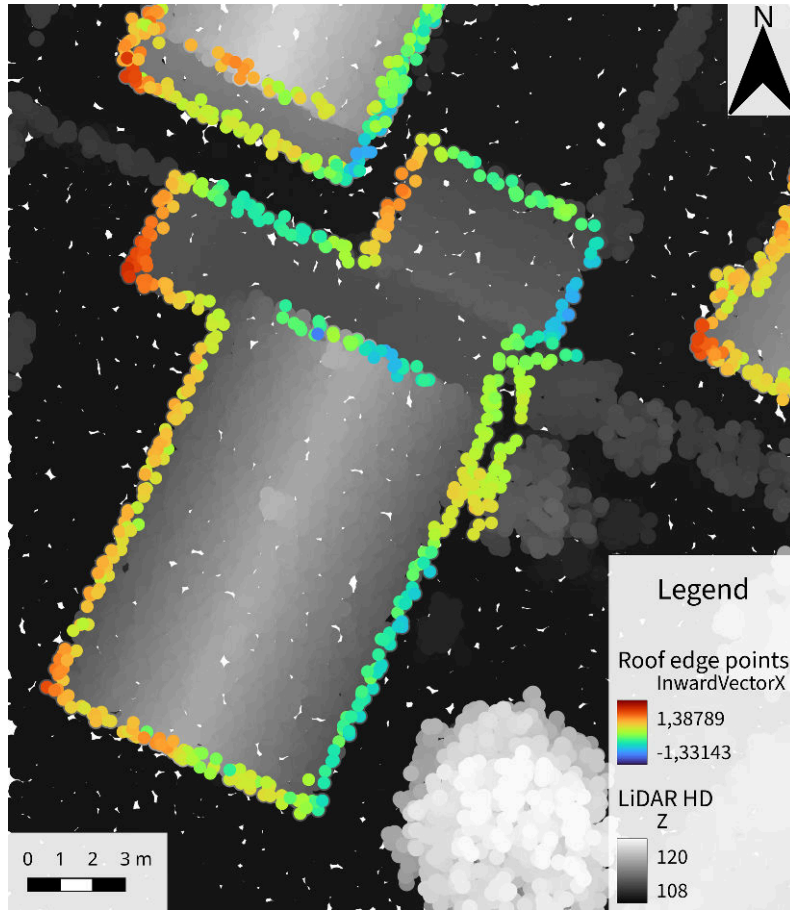
(a) Coloured by classification.



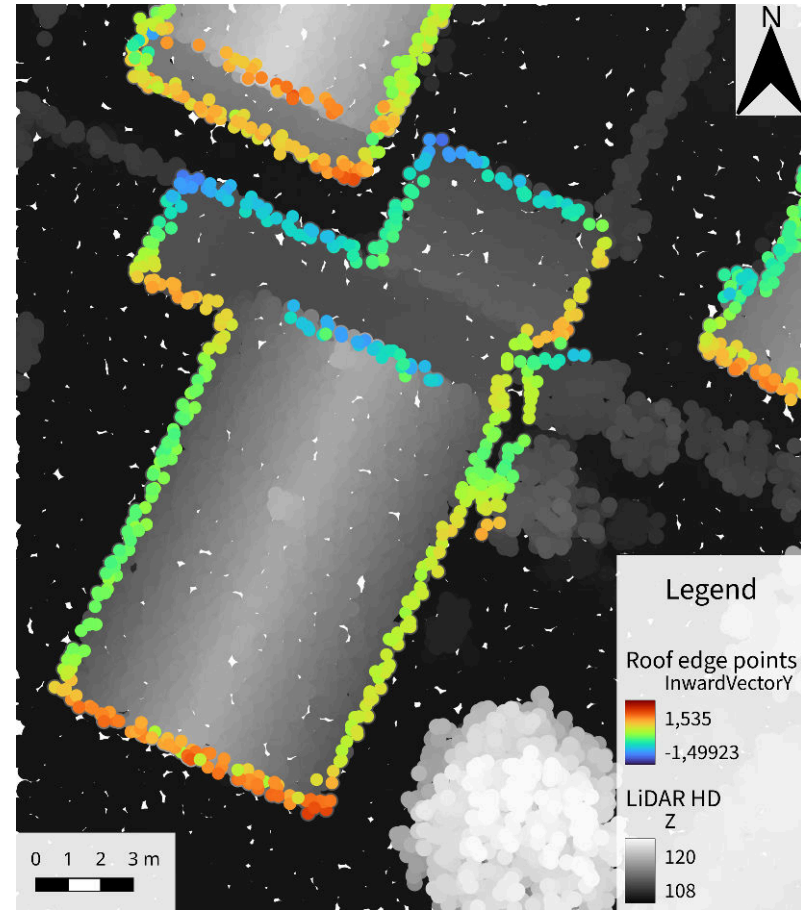
(b) Coloured by height.

Figure 10: The LiDAR HD data.

Illustration with a real building



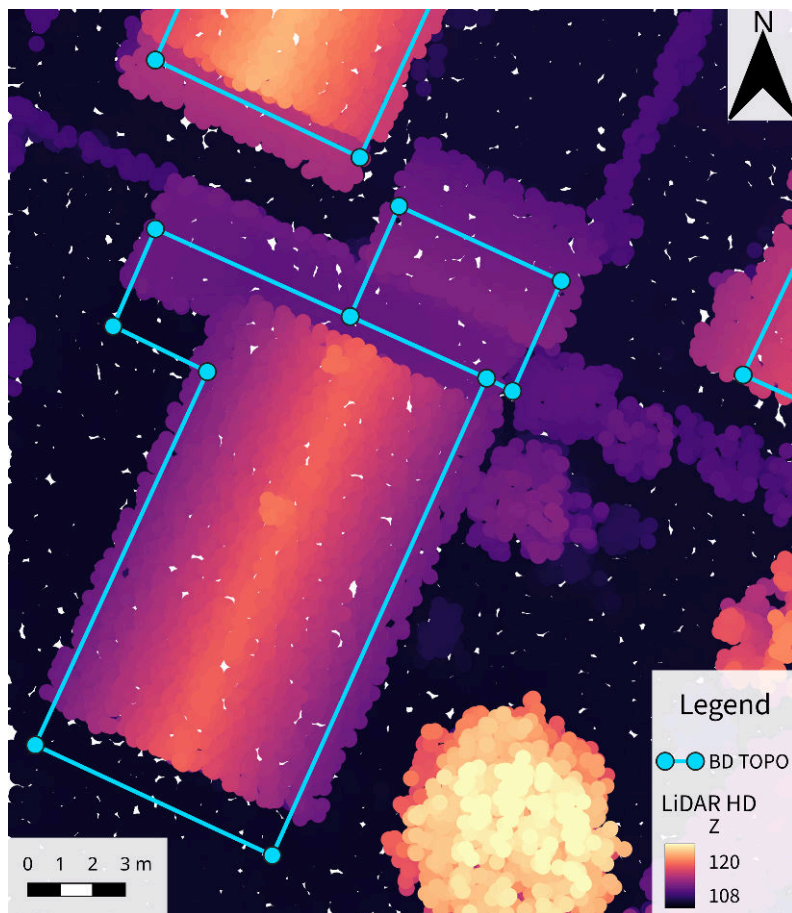
(a) Coloured by the X component.



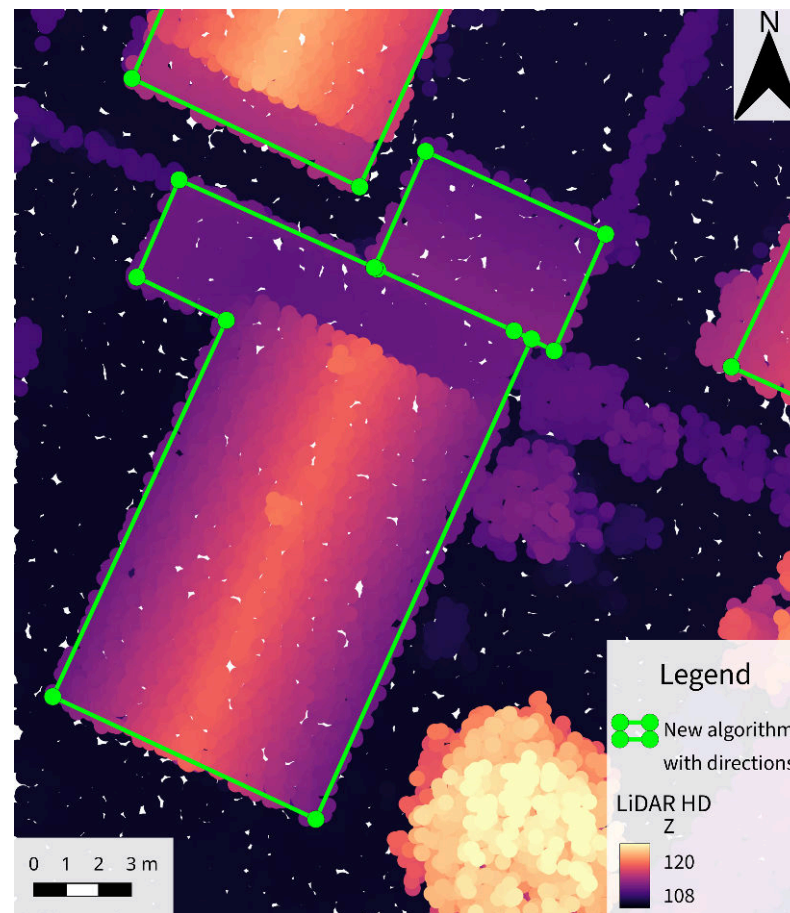
(b) Coloured by the Y component.

Figure 11: The detected roof edge points coloured by their inward vector.

Illustration with a real building



(a) Initial outlines in the BD TOPO.



(b) Final roofprint produced by the algorithm.

Figure 12: Comparison of BD TOPO outlines and roofprints computed with our algorithm.

Footprints



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Footprints evidence

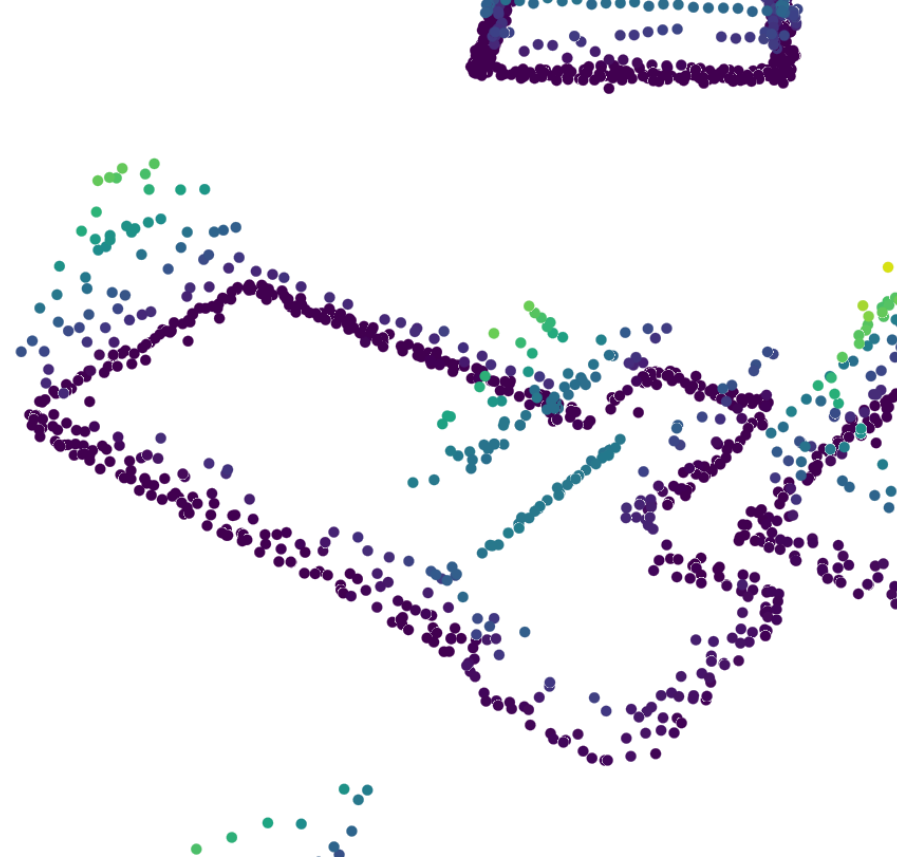
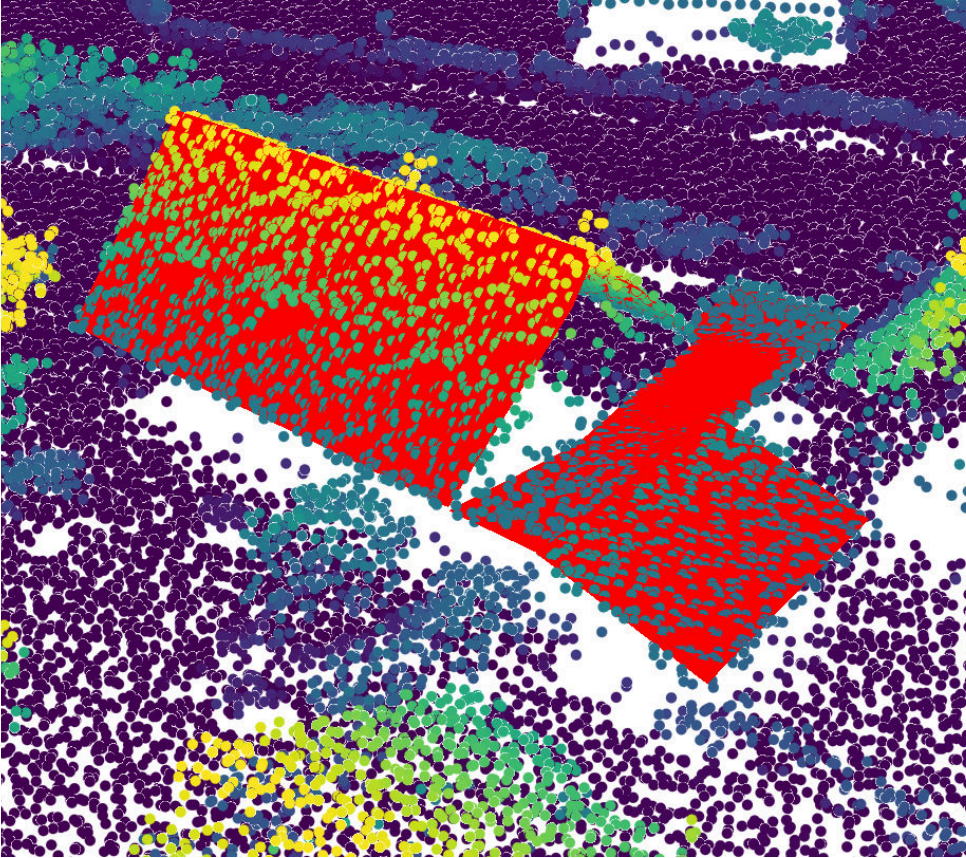


Figure 13: Illustration of the 3D roof structure and the selected points for the footprint computation.

Illustration with a real building

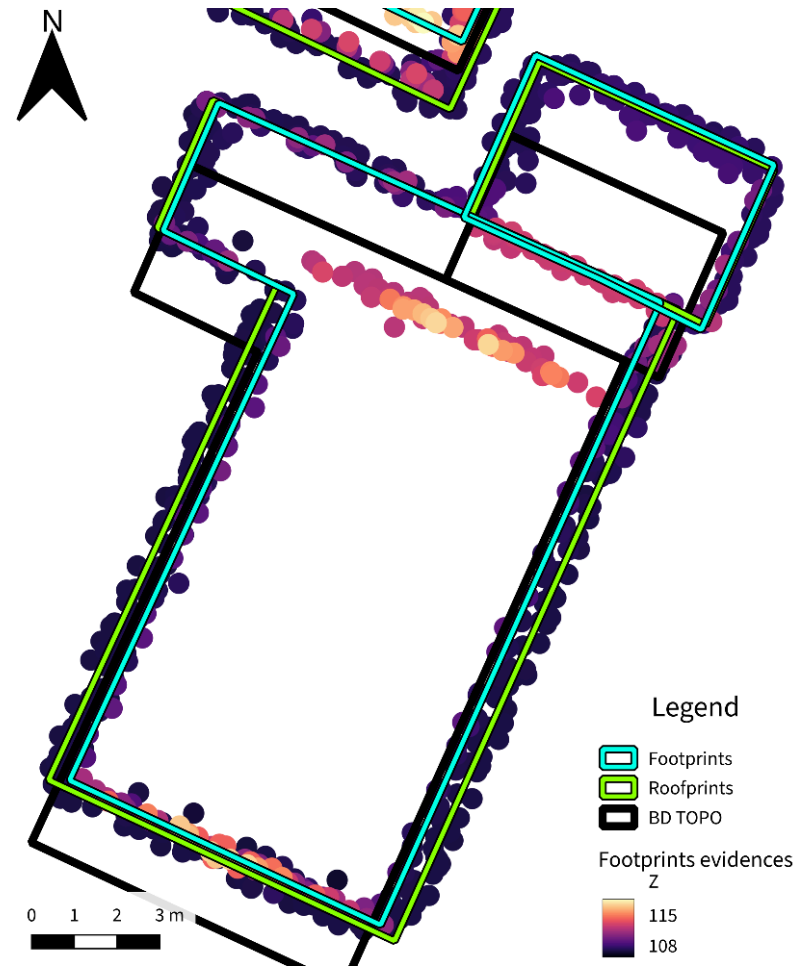
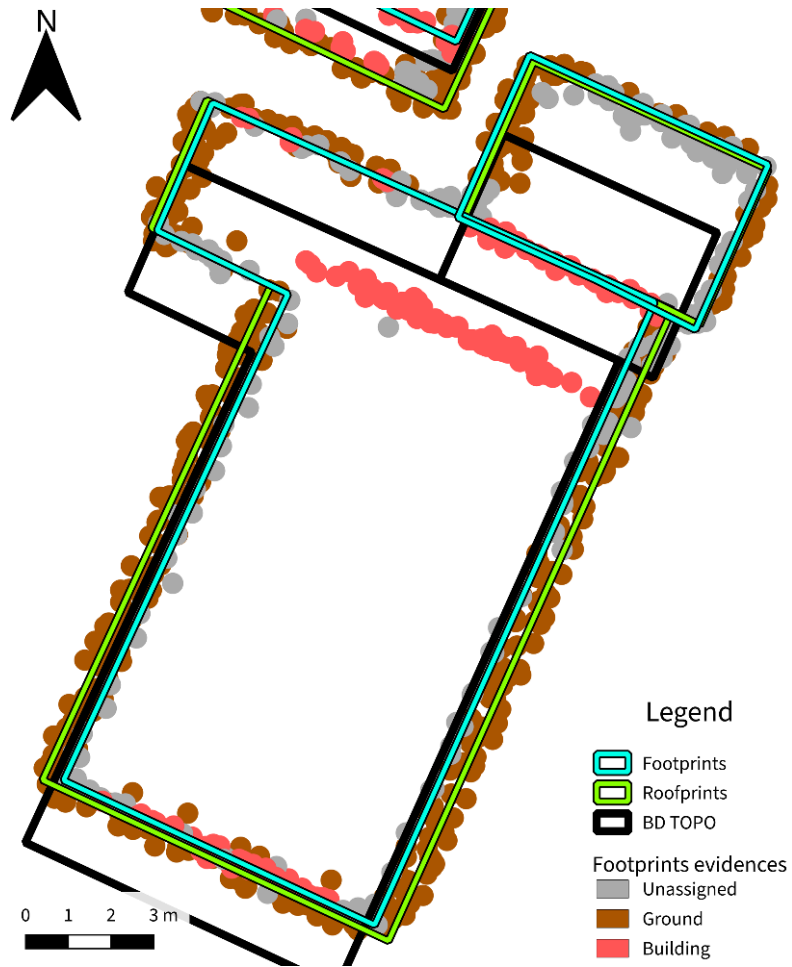


Figure 14: Illustration of the computed footprints.

Illustration with a real building

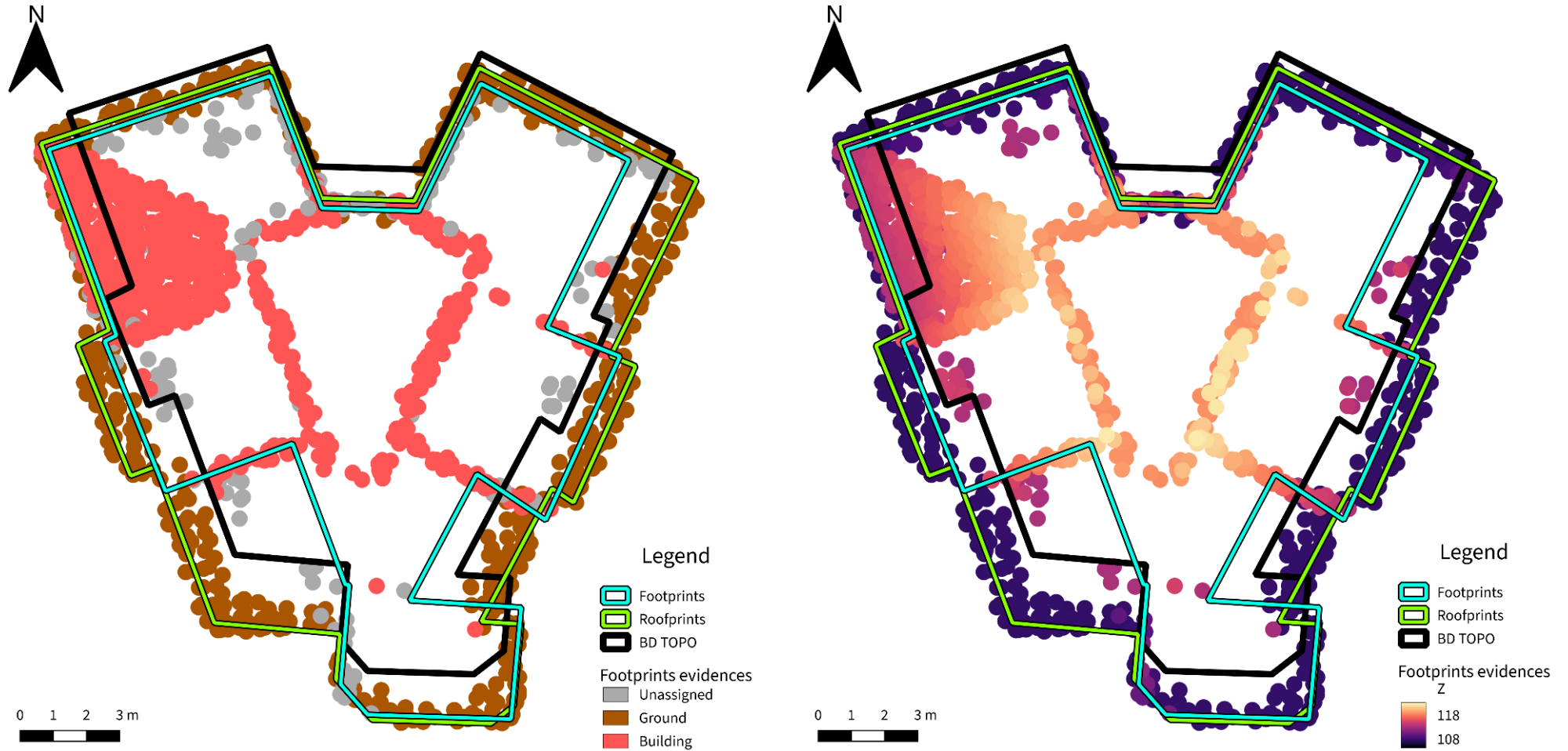


Figure 15: Illustration of the computed footprints.

Conclusion

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Result of the thesis	Benefit
Roofprint and footprint	More precise computations with 2D data LoD2.2 → LoD2.3
Better georeferencing	Better visualisation in context Better 3D roof models

Thank you for your attention!

Link to the slides:

https://alexandre-bry.github.io/MSc_Thesis-Report/slides_pages/2026_06_26-A4.html



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References

- Biljecki, F., Ledoux, H., & Stoter, J. (2016). An improved LOD specification for 3D building models. *Computers, Environment and Urban Systems*, 59, 25–37. <https://doi.org/10.1016/j.compenvurbsys.2016.04.005>
- Wu, T., Vallet, B., Brédif, M., Caraffa, L., & Vosselman, G. (2026). *Reverse engineering LiDAR pulse emission points*.

Appendix

Context	1
Polygon deformation	8
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Results

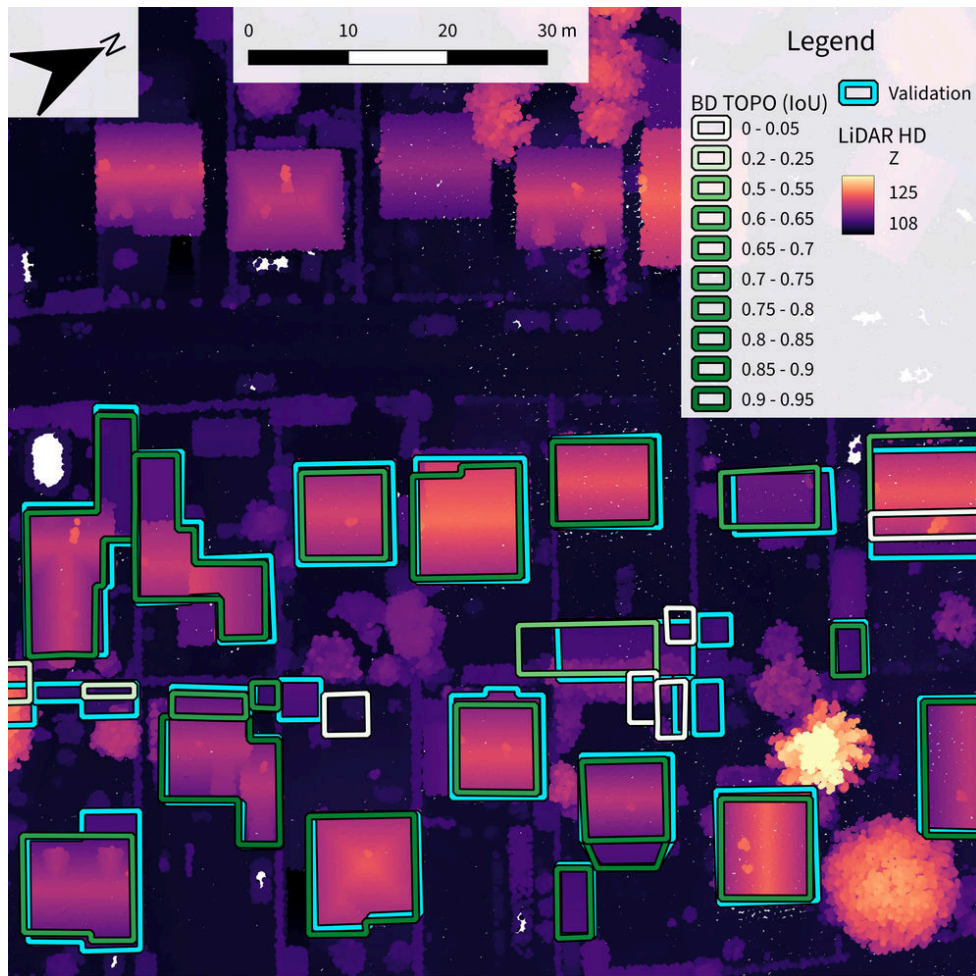


Figure 16: BD TOPO.

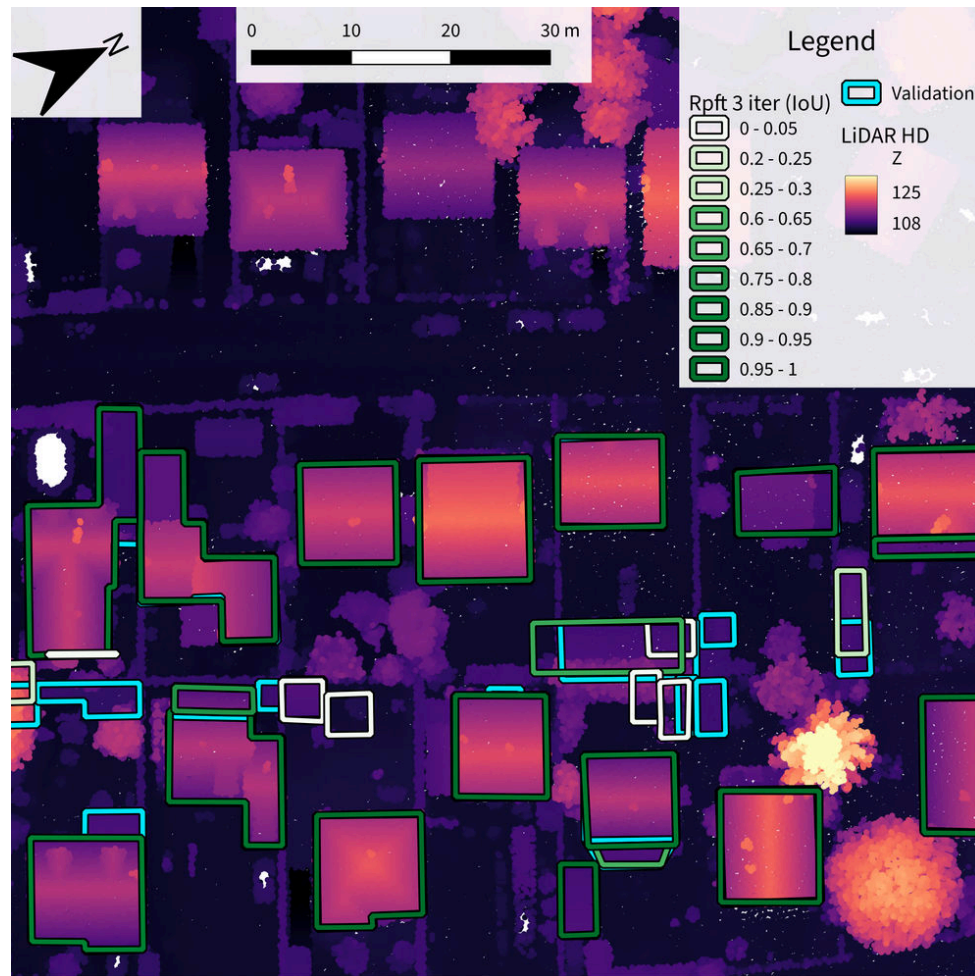


Figure 17: Roofprints after 3 iterations.

Results

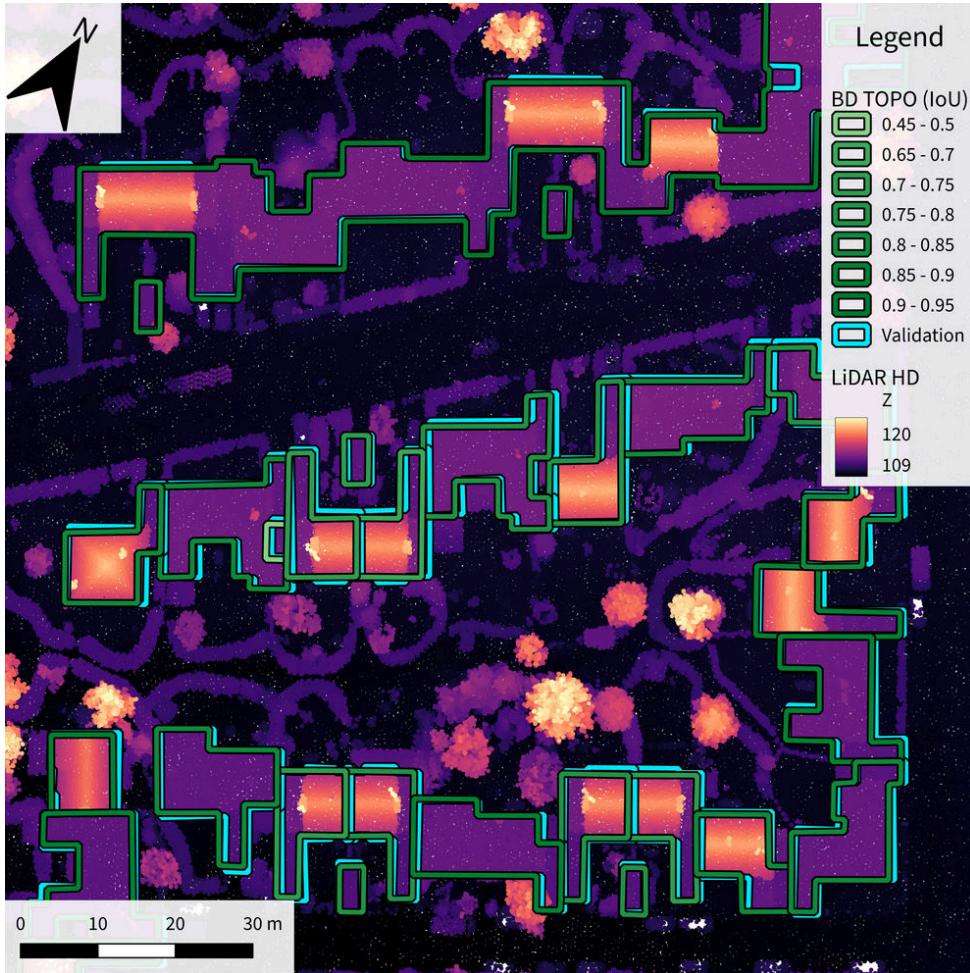


Figure 18: BD TOPO.

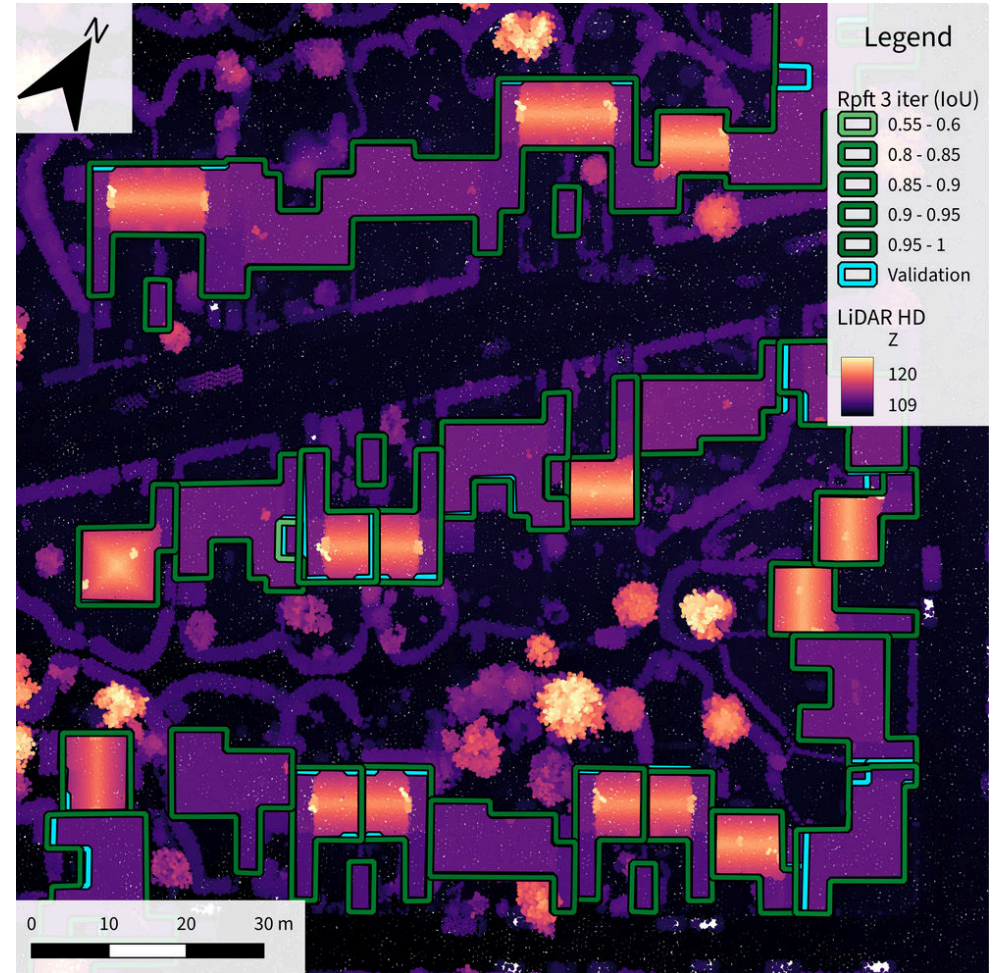


Figure 19: Roofprints after 3 iterations.

Results

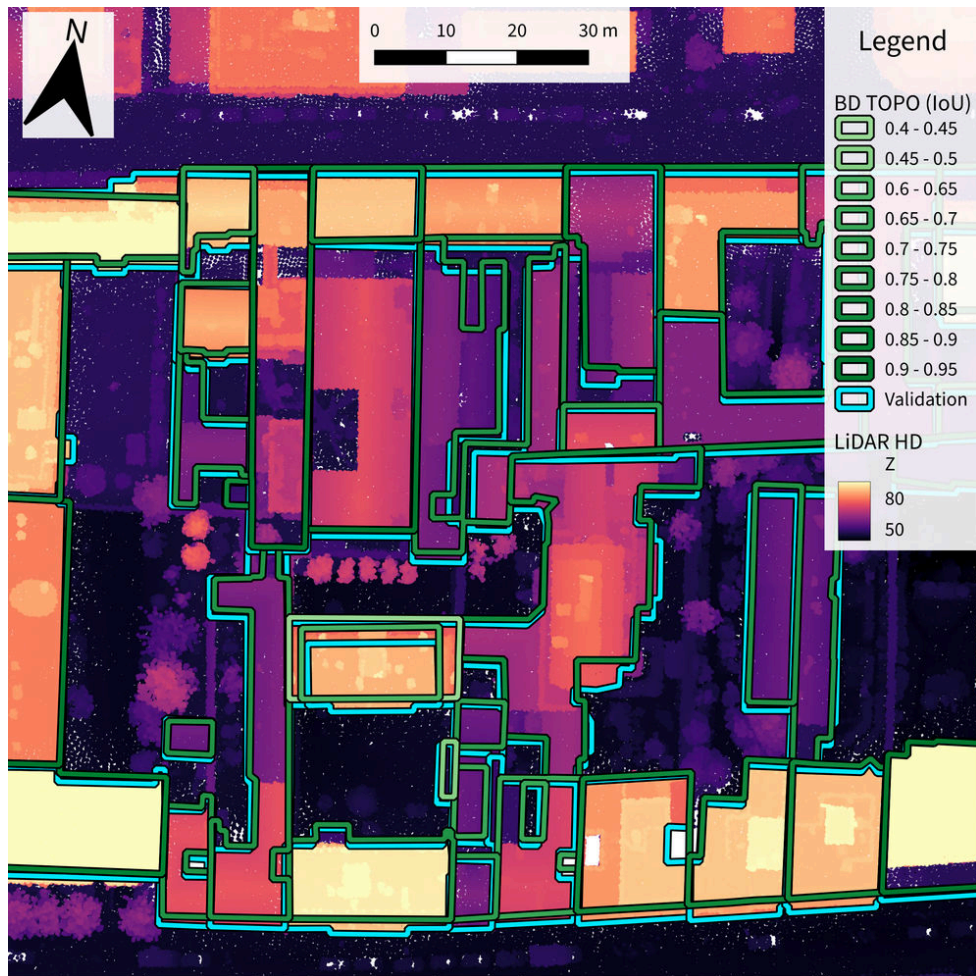


Figure 20: BD TOPO.



Figure 21: Roofprints after 3 iterations.

Overview of the methodology

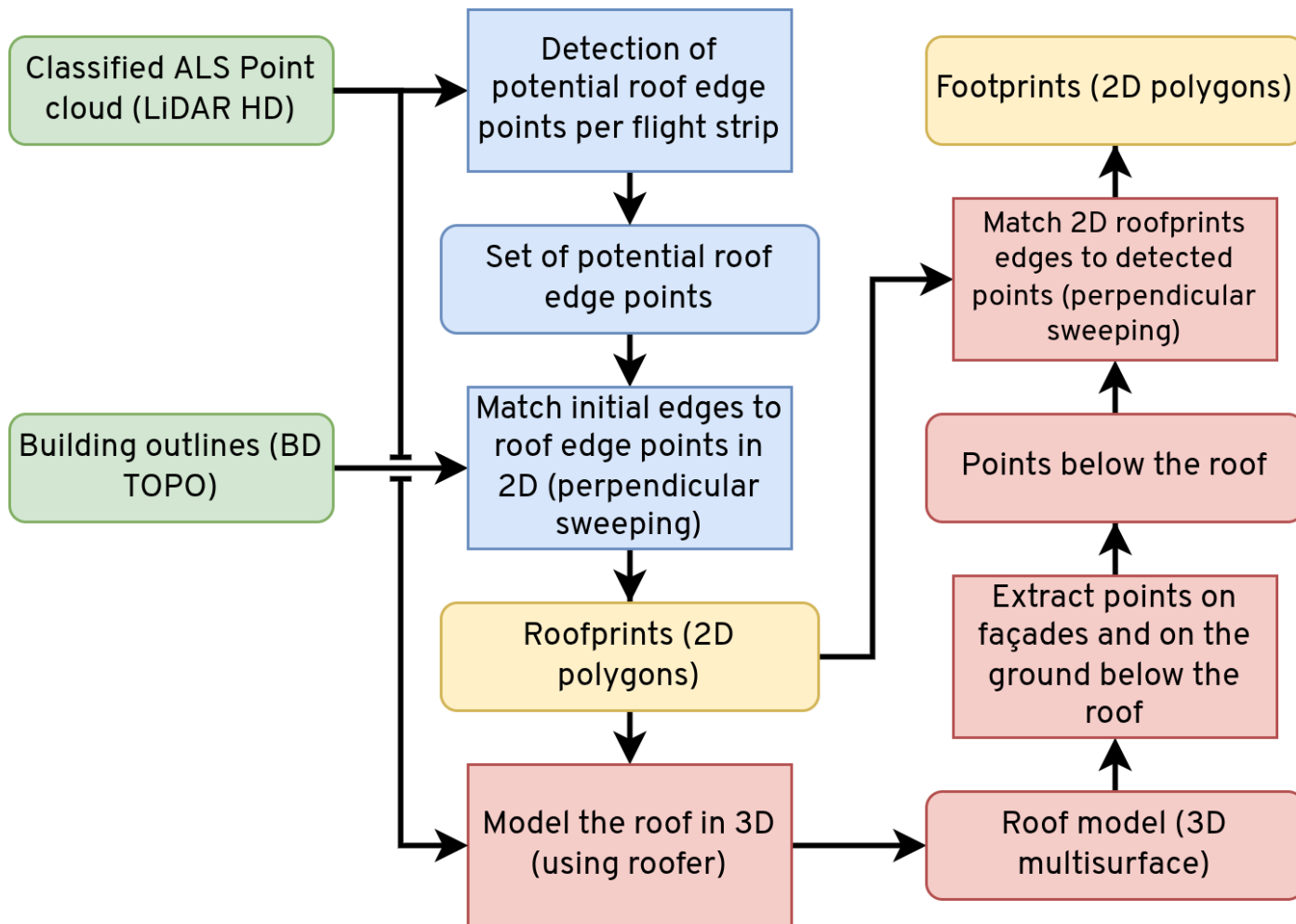


Figure 22: Overview of the pipeline.